Edge-Guided Refinement Network With Similarity Perception for Salient Object Detection in Optical Remote Sensing Images

Xin Wang, Xian Fang, *Member, IEEE*, Mingfeng Jiang, Jinchao Zhu, and Zhigao Li

***Abstract*—Salient object detection in optical remote sensing image (ORSI-SOD) aims to segment salient regions from high-resolution remote sensing images. However, most existing ORSI-SOD methods primarily rely on feature learning of regions to address the issue of blurred edges, while neglecting the potential advantages of similarity calculation in inferring edge clues. To overcome this issue, we propose a novel network with similarity perception termed the edge-guided refinement network (ERNet), which distinguishes the edge region of salient objects from coarse to fine through two-stage similarity calculation. First, we introduce the adaptive uncertainty calibration module (AUCM), which utilizes the proposed similarity-based edge perception mechanism (SEPM) to adaptively calibrate edge uncertain information. Second, to effectively capture global semantic information, we propose the hierarchical semantic reconstruction module (HSRM), which comprehensively correlates different levels of semantic clues from both internal and external perspectives. Finally, to supplement the local detail of salient objects, we design the dynamic detail interaction module (DDIM), which dynamically extracts detailed information of objects at different scales. Extensive experiments on three challenging benchmark datasets have demonstrated the remarkable superiority of our ERNet compared to 30 state-of-the-art models. The source codes are publicly available at https://github.com/xinwang11/ERNet**

***Index Terms*—Detailed interaction, edge guidance, optical remote sensing image (ORSI), salient object detection (SOD), semantic reconstruction, similarity perception.**

I. Introduction

SALIENT object detection (SOD) is a fundamental task of computer vision that detects the most visually appealing objects or regions from given complex scenes [1]. Due to the excellent generality and scalability of SOD, it has been widely applied in the preprocessing stage of many tasks, such as image segmentation [2], [3], surface defect detection [4], [5], change detection [6], [7], and object tracking [8], [9]. Recently, with the rapid development of applications such as environmental monitoring and military reconnaissance, many researchers have begun to explore SOD methods in optical remote sensing image. However, significantly different from manually captured natural scene images (NSIs), ORSIs are typically collected in space or high altitude using devices with optical sensors such as drones or satellites, which result in ORSIs having higher resolution and more complex backgrounds. Therefore, directly applying existing NSI detection methods to ORSI may not be appropriate.

In recent years, researchers have proposed many effective models based on convolutional neural network (CNN) or transformer, which have greatly promoted the development of salient object detection in optical remote sensing image (ORSI-SOD). It is worth noting that some methods have achieved encouraging results by focusing on the edge information of objects. For example, both ERPNet [10] and BAFSNet [11] designed the parallel decoding network, which additionally includes a decoder specifically for edge detection of salient objects to improve the edge details of salient objects. AESINet [12] proposed the difference perception mechanism, which adaptively perceives the edge of salient objects by calculating the value between the average and maximum of the pixels. Although the above methods have achieved some achievements in processing edge information, the relatively simple design in extracting edge features may bring some limitations. First, the size variation of salient objects in ORSIs is more extreme compared to regular NSIs. Faced with this challenge, the edge decoders of ERPNet and BAFSNet rely solely on generic fundamental computational components, which lack specific optimization for ORSI scenarios. Second, salient objects such as lakes and islands in ORSIs often have severely uneven edges. Although AESINet can detect the edge of salient objects by calculating the difference between average pooling and maximum pooling of edge pixels, this method is unable to accurately express the edge features when the difference is too small.

Different from other methods, we consider imitating human behavior to distinguish the edge of salient objects, as shown in Fig. 1. In Fig. 1(a), humans usually first locate the edge region of salient objects and then focus on this region to subdivide the edge of objects. Correspondingly, in Fig. 1(b), the proposed similarity-based edge perception mechanism (SEPM) first locates the coarse edge of salient objects through large-scale similarity calculation and gating, and then subdivides the fine edge through small-scale similarity calculation and gating. Compared to the human behavior of localization and

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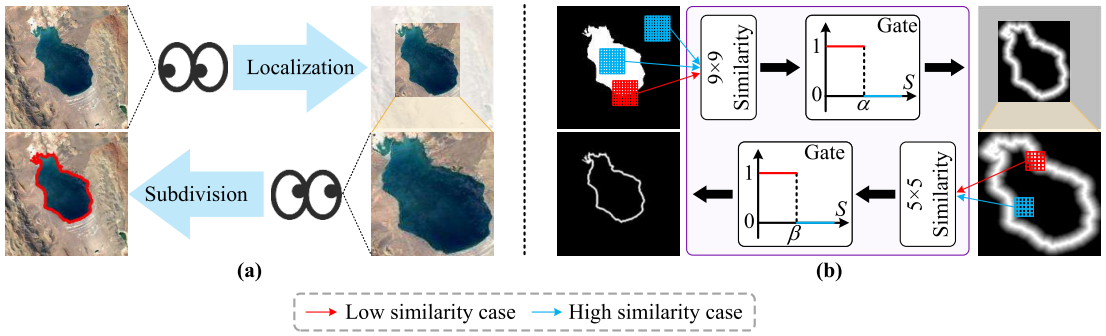


Fig. 1. Main idea of our proposed SEPM. (a) Way humans identify the edge of salient objects. (b) Imitating human behavior through similarity calculation and gating.

subdivision, SEPM contributes to adaptive inference of edge cues through two-stage similarity perception.

Inspired by the above intuitive ideas, we propose a novel edge-guided refinement network (ERNet), which utilizes cosine similarity calculation to simulate the process of human capturing object edge. Specifically, to effectively calibrate the uncertain information in the edge region of salient objects, we design the adaptive uncertainty calibration module (AUCM). The core of AUCM lies in the proposed SEPM, which adaptively obtains the edge information of salient objects through two-stage similarity calculation at different scales. To localize salient objects more precisely in complex backgrounds, we develop the hierarchical semantic reconstruction module (HSRM). This module focuses on the semantic features generated by high-level encoders, which hierarchically reconstruct deep semantic clues to improve the discriminative ability of the model. To suppress the interference of noise, we develop the dynamic detail interaction module (DDIM). This module focuses on the detailed features generated by low-level encoders, which employ dynamic interaction and progressive aggregation to enhance the representation of local details.

Our main contributions can be summarized in the following four aspects.

1) We propose a novel ERNet, which constructs full-stage edge guidance with pixel similarity and hierarchically refines detailed and semantic information at different stages. To the best of authors' knowledge, the proposed network is the first to design the edge decoder in a two-stage pixel similarity refinement manner for ORSI-SOD.
2) We design the AUCM, which adaptively distinguishes uncertain regions between foreground and background through the SEPM. This mechanism calculates pixel similarity at different scales from coarse to fine to enhance the edge information of salient objects.
3) We develop the HSRM, which reconstructs semantic features through internal and external correlations to assist the model in accurately capturing the exact location of salient objects.
4) We develop the DDIM, which dynamically weights local features at different scales and progressively aggregates cross-scale features via a central propagation strategy to enrich the detailed information of salient objects.

II. Related Work

A. SOD in NSIs

In the early stages of SOD, researchers mainly conducted a series of explorations on NSIs based on handcrafted features and saliency priors [13]. Most early methods designed classification models by extracting interpretable clues in an unsupervised manner, which often failed to accurately detect salient objects in scenes. Nowadays, with the rapid advancement of convolutional neural networks, many SOD models based on encoder–decoder architectures have been proposed. Due to the successive publication of the SOD dataset, researchers have explored various fully supervised detection methods. For example, Wang et al. [14] designed a multilevel mixed loss at the pixel level, region level, and object level to enhance salient object features. Li et al. [15] proposed a novel adaptive neighborhoodwise graph convolution to model global contextual relationships. Furthermore, some researchers have made some efforts in weakly supervised methods that are less dependent on the dataset. Cong et al. [16] corrected coarse labels by adding a label refinement branch on top of the SOD branch. Piao et al. [17] proposed the noise-sensitive training strategy to better utilize pseudo-labeled samples. In addition, lightweight detection networks have also attracted the attention of many researchers. Shen et al. [18] designed the single-scale detection algorithm that effectively reduces the computational cost of the model. Wang et al. [19] proposed the lightweight saliency backbone without pretraining to significantly reduce feature redundancy. Shen et al. [20] introduced the mini-region of interest to obviously reduce the computational complexity of feature extraction.

Although the methods mentioned above have shown promising performance, considering the inherent differences between ORSIs and NSIs, developing a framework adapted for ORSIs to mitigate the disparities in object size and texture is essential.

B. SOD in ORSIs

As an influential branch of SOD, ORSI-SOD has attracted increasing attention from researchers. The early classic ORSI-SOD methods mainly focused on manually known clues. Zhang et al. [21] adopted the low-rank matrix recovery method to adaptively weight and fuse multiple feature channels. Wang et al. [22] combined spatial and spectral information to

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adaptively highlight salient objects. Ma et al. [23] generated high-quality saliency maps through superpixel-level sampling and pixel-level mapping. However, these traditional methods usually have huge computational cost, which fails to satisfy the needs of handling a large number of ORSIs.

Recently, many deep learning-based methods have been emerging in the ORSI-SOD field. Early deep learning methods mainly focused on a CNN-based network framework. For instance, Li et al. [24] proposed a multicontent complementation network, which enhances salient regions by fusing and supplementing various types of features such as foreground and background features with each other. Song et al. [25] fused adjacent features of the encoder to improve the performance of model. Zhai et al. [26] combined multiscale and edge information of features to assist the network in generating fine saliency maps. Due to the increasing popularity of transformer-based network framework in recent years, newly proposed methods are beginning to be designed based on it. Li et al. [27] first proposed designing the ORSI-SOD method on the transformer-based network framework. Gao et al. [28] designed an adaptive spatial tokenization encoder and three cascaded decoders to extract and aggregate global dependencies. Han et al. [29] significantly reduced the complexity of the model through dynamic region-aware and multipath enhancement. Furthermore, some researchers introduce a novel paradigm based on a diffusion model. Han et al. [30] designed a denoising network, which effectively enhances the refinement ability of the model through the proposed consistency assessment strategy.

The above approaches provide many innovative theoretical foundations for ORSI-SOD. However, most models pay little or no attention to edge information, which may be disadvantageous to handle ORSIs with highly similar foreground and background. Consequently, we propose a novel edge-guided network, which designs the edge decoder based on similarity from a new perspective to perceive edge information.

C. Edge Guidance Architecture

Edge guidance is widely used in many computer vision tasks and has demonstrated excellent performance. In general, edge-guided networks mainly rely on edge map supervision or edge loss to optimize the edge information of objects. Zhao et al. [31] accomplished SOD by complementing and coupling salient information and edge information. Sun et al. [32] designed the edge generator to alleviate the edge loss issue of the model in RGB-depth scenes. Zhou et al. [33] introduced the prior edge maps generated by two modalities to guide the model in completing RGB-thermal scene parsing. Yang et al. [34] established edge branching and guidance modules to extract buildings from remote sensing images. Sun et al. [35] detected camouflaged objects by integrating edge information from coarse to fine and supervising it. In the field of ORSI-SOD, Wang et al. [36] proposed the interactive guidance loss for joint learning of edge information. Tu et al. [37] designed two serial decoders to continuously refine edge features. Zhou et al. [38] explicitly and implicitly supplemented edge information through edge modules and loss functions.

However, most edge-guided methods focus on feature learning of regions to distinguish edge information, which makes it difficult for them to completely capture and utilize the unique characteristics of edges. Different from these methods, our edge decoder accurately perceives edge features from coarse to fine by leveraging pixel similarity at different scales.

III. Methodology

A. Overview

The overall architecture of our proposed ERNet is shown in Fig. 2. In general, the model consists of four important components, including an encoder for extracting basic features, an edge decoder for edge-guided supervision, two modules for semantic and detailed feature enhancement, and a decoder for saliency prediction. Specifically, we input the image into the pyramid vision transformer (PVT) to extract the global dependencies. PVT contains four transformer encoder blocks, which can generate four basic encoder features denoted as $E^i \in \mathbb{R}^{N\times C\times H\times W}(i = 1, 2, 3, \text{ and } 4)$. These encoder features are then fed into the edge guidance branch and the saliency prediction branch, respectively. In the edge guidance branch, the number of channels for encoder features is first unified to 128 through the mapping function to improve computational efficiency. These features can be further represented as $E_m^i \in \mathbb{R}^{N\times 128\times H\times W}$. Subsequently, features E_m^i are input into the edge decoder consisting of three cascaded AUCMs to adaptively distinguish salient object edges and generate the edge prediction map $F_{\text{edg}} \in \mathbb{R}^{N\times 1\times H\times W}$. In the saliency prediction branch, to equip the model with a deeper understanding of the semantic context, we feed the higher level encoder features $\{E^2, E^3\}$ and $\{E^3, E^4\}$ into HSRM, which generates the enhanced semantic features F_S^3 and F_S^4. In addition, we feed the lower level encoder features $\{E^1, E^2\}$ and $\{E^2, E^3\}$ into DDIM to adequately integrate detailed features of different scales, which further generates the enhanced detailed features F_D^1 and F_D^2. Finally, the backbone decoder generates a high-quality saliency prediction map $F_{\text{sal}} \in \mathbb{R}^{N\times 1\times H\times W}$ by sufficiently merging semantic, detail, and edge information. Another point to mention is that the saliency prediction map F_{sal} and edge prediction map F_{edg} are supervised separately from their corresponding ground truth.

B. Adaptive Uncertainty Calibration Module

For the ORSI-SOD task, edge representation that includes complete and discriminative spatial structures can effectively alleviate the problem of edge blurring. As is well-known, the high-level features of the encoder contain global context, while the low-level features of the encoder contain local context. Although some existing methods [12] believe that local context contributes to supplement detailed information, there is also evidence that global context is beneficial for locating the overall shape of objects [39]. Therefore, to thoroughly exploit edge information, we establish the full-stage edge decoder with the proposed cascaded AUCM to extract edge detail.

The detailed structure of AUCM is illustrated on the right side of Fig. 2. For simplicity, we assume that the two inputs of ith AUCM are F_l and F_h, where F_l is the feature map

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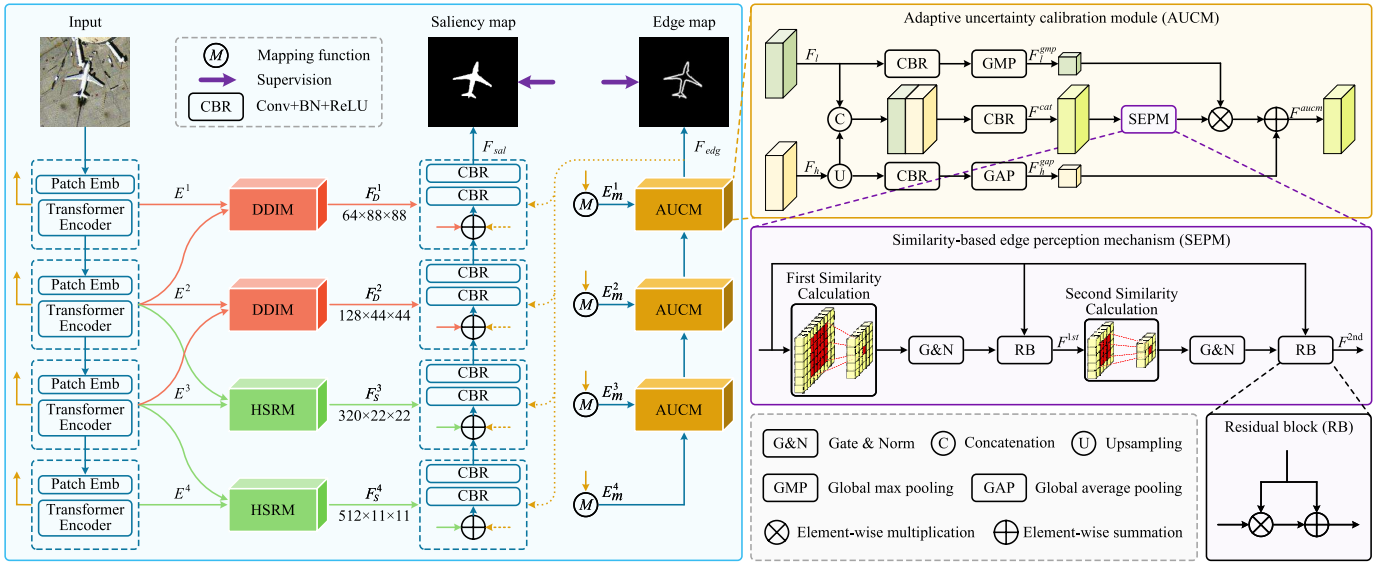


Fig. 2. Architecture of the proposed ERNet. The network mainly consists of saliency prediction branch and edge guidance branch. The former embeds DDIM and HSRM in the skip connections to refine local detail and global semantic, while the latter extracts edge information with pixel similarity and guides saliency prediction branch through the edge decoder composed of cascaded AUCM.

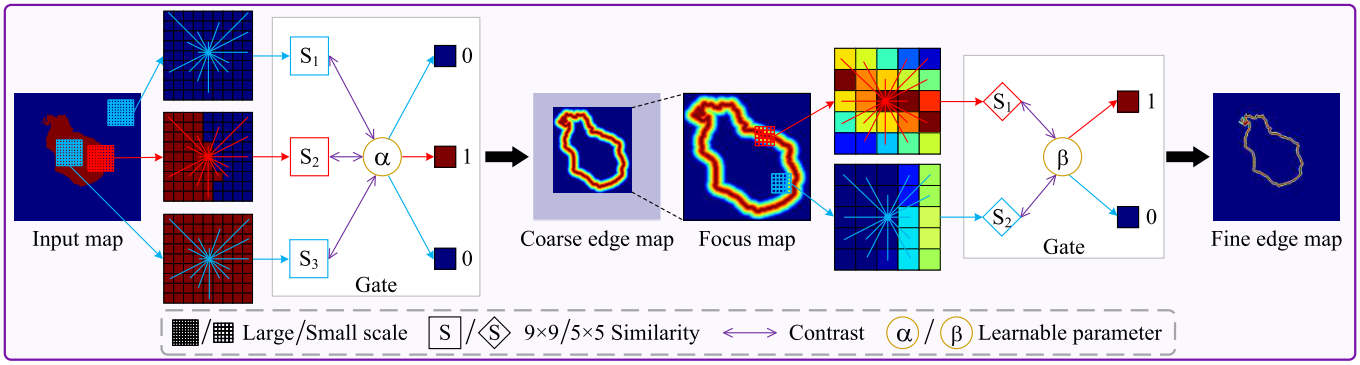


Fig. 3. Illustration of SEPM. In these two stages, this mechanism generates coarse and fine edge maps through similarity calculation from large to small scales and gating mechanisms with learnable thresholds of α and β.

obtained by mapping the ith layer features of the encoder, and F_h is the feature map obtained by mapping the $i+1$th layer features of the encoder or the output feature map of the $i+1$th AUCM. Since F_l and F_h are inconsistent in spatial dimensions, we first upsample F_h to align the spatial scale. Then, we concatenate these two features to form a new fusion branch and equip all branches with convolutional blocks to enhance the feature representation. Furthermore, for the initial two branches, we employ global average pooling and global max pooling to extract their global guidance information. The above process can be expressed as follows:

$$F_l^{\text{gmp}} = \text{GMP}\left(\text{CBR}\left(F_l\right)\right) \tag{1}$$

$$F^{\text{cat}} = \text{CBR}\left(\text{Cat}\left(F_l, UP\left(F_h\right)\right)\right) \tag{2}$$

$$F_h^{\text{gap}} = \text{GAP}\left(\text{CBR}\left(UP\left(F_h\right)\right)\right) \tag{3}$$

where GAP and GMP represent the global average pooling and global max pooling, respectively, UP and Cat denote the upsampling and concatenation operations, respectively, and CBR means the combination of a convolution layer, batch normalization, and rectified linear unit (ReLU).

After the above procedures, we adopt SEPM for the feature maps of the fusion branch to adaptively extract edge information. Our choice to adopt cosine similarity for calculating pixel similarity in small regions is attributed to its benefit in accurately capturing the changing trends between pixels and its low sensitivity to the amplitude difference of pixel values. First, we introduce the pixel-level cosine similarity and the gating mechanism as follows:

$$S^{i,j}(X) = \frac{\sum_{c=1}^{C} X_{c,h,w} \cdot X_{c,h+i,w+j}}{\sqrt{\sum_{c=1}^{C}\left(X_{c,h,w}\right)^2}\sqrt{\sum_{c=1}^{C}\left(X_{c,h+i,w+j}\right)^2}} \tag{4}$$

$$\text{Gate}^x(X) = \begin{cases} 1, & X < x \\ 0, & X > x \end{cases} \tag{5}$$

where $S^{i,j}(X) \in \mathbb{R}^{(i\times j-1)\times H\times W}$ contains the cosine similarity between each pixel and its $(i\times j-1)$ neighbor pixels. Afterward, we obtain the fine edge perception map by performing twice similarity calculations at different scales and gating with different thresholds on the input feature map. We express the

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above process as follows:

$$F^{1st} = \mathrm{RB}(L2_{\mathrm{norm}}(\mathrm{Gate}^{\alpha}(S^{9,9}(F^{\mathrm{cat}})))) \quad (6)$$

$$F^{2nd} = \mathrm{RB}(L2_{\mathrm{norm}}(\mathrm{Gate}^{\beta}(S^{5,5}(F^{1st})))) \quad (7)$$

where RB denotes the residual block, $L2_{\mathrm{norm}}$ represents the calculation of $L2$-norm for the channel dimension, α and β are two different learnable parameters that are initialized to 0.3 and gradually converge to appropriate weights, F^{1st} is the coarse edge map obtained after first similarity calculation, and F^{2nd} is the fine edge map obtained after second similarity calculation.

The more detailed explanation of SEPM is shown in Fig. 3. In the feature space, edges are characterized by drastic changes in neighboring feature vectors, which typically correspond to low pixel similarity. Since the severely irregular shape of object edges, we gradually locate and subdivide the edges through a two-stage manner to acquire edge feature maps from coarse to fine. In the first stage, we utilize the 9×9 similarity calculation kernel to scan the entire map. After calculating the similarity of the entire map, we adopt the learnable gating mechanism to assign low confidence to salient and nonsalient regions and high confidence to edge regions, which generates the coarse edge map. In the second stage, we use a smaller 5×5 similarity calculation kernel to scan the coarse edge map generated in the first stage, which further refines the features of the edge regions through gating mechanisms with different thresholds and generates the final fine edge map.

Finally, we guide and complement the fine edge map F^{2nd} obtained from the fusion branch with the global information F_l^{gmp} and F_h^{gap} from the original two branches, and then obtain the output of AUCM as follows:

$$F^{\mathrm{aucm}} = \left(F^{2nd} \otimes F_l^{\mathrm{gmp}}\right) \oplus F_h^{\mathrm{gap}} \quad (8)$$

where $\otimes$ means the elementwise multiplication, and $\oplus$ means the elementwise summation.

In summary, our AUCM assists the model in learning discriminative representations to calibrate local uncertainty through three-branch interaction and continuously refines edge information with the embedded SEPM.

C. Hierarchical Semantic Reconstruction Module

Effectively exploiting semantic information from high-level encoder features contributes to locating salient objects of various shapes and scales. However, some existing methods [36], [40] mainly aggregate semantic information in the interfeature or intrafeature dimension, while ignoring the collaborative learning of these two dimensions. Therefore, we propose the HSRM to comprehensively enhance the coherence of global semantic from both internal and external perspectives.

The structure of HSRM is illustrated in Fig. 4, which mainly consists of internal correlation block (ICB) and external correlation block (ECB). Different from the modality-aware dynamic aggregation module (MDAM) in ConTriNet [41], our ICB and ECB mine deep semantic information from both internal and external features through global–local collaboration and progressive refinement. It is worth mentioning that ICB focuses on internal enhancement of feature, while ECB focuses on external fusion of features. Specifically, given the two inputs E^{i-1} and E^{i} of the ith HSRM, we first employ pixel unshuffle [42] and convolution block to align the spatial and channel dimensions for the low-level feature E^{i-1}, and employ two different convolution blocks for the high-level feature E^{i} to construct the dual-branch as follows:

$$F^{i-1} = \mathrm{CBR}\left(\mathrm{PU}\left(E^{i-1}\right)\right) \quad (9)$$

$$F_c^{i} = \mathrm{CBR}\left(E^{i}\right) \quad (10)$$

$$F_d^{i} = \mathrm{DBR}\left(E^{i}\right) \quad (11)$$

where DBR denotes the combination of dilated convolution, batch normalization, and ReLU, and PU denotes the pixel unshuffle operation. The dual-branch architecture enables the purification of relevant semantic clues from the high-level feature E^{i} while suppressing background semantic interference.

1) *ICB:* It adopts the combination of global and local methods to focus on the semantic information within the high-level feature. As shown in the left part of Fig. 4, given the inputs F_c^i and F_d^i, we first fuse the two features and apply global average pooling to extract joint global clues from the concatenated feature. Subsequently, we generate the new activation map F_{sig}^i through double fully connected layer mapping as follows:

$$F_{\mathrm{sig}}^{i} = \mathrm{FC}_S(\mathrm{FC}_{\mathrm{BR}}(\mathrm{GAP}(\mathrm{Cat}(F_c^i, F_d^i)))) \quad (12)$$

where FC_S denotes the fully connected layer with sigmoid function, and $\mathrm{FC}_{\mathrm{BR}}$ denotes the fully connected layer with batch normalization and ReLU function. Next, we divide the feature F_d^i into two equal parts along the H- and W-directions to obtain four feature vectors and locally weight them using channel attention and spatial attention. Subsequently, we recombine the weighted features to obtain the local potential features. In addition, for another feature F_c^i, we directly mine the global potential features through channel attention and spatial attention. Finally, we quantify the semantic importance of local and global potential features separately through the activation map F_{sig}^i and its derived map, which enables us to filter out the most representative semantic information within the feature. The above process can be expressed as

$$F^{i} = \mathrm{Cat}\left(\mathrm{CS}\left(F_c^i\right) \otimes \left(1 - F_{\mathrm{sig}}^i\right), \quad \mathrm{RCS}\left(F_d^i\right) \otimes F_{\mathrm{sig}}^i\right) \quad (13)$$

where CS represents the combination of channel attention and spatial attention, and RCS adds the feature reconstruction operation based on CS.

2) *ECB:* To further refine the semantic interaction between high-level features and reduce noise contamination, ECB reconstructs the feature generated by ICB thoroughly and fuses it with another high-level feature, which is expected to alleviate the problem of semantic information incompatibility between different layers to some extent. As depicted in the right part of Fig. 4, we first apply two convolutional blocks to the inputs F^{i-1} and F^{i} to optimize the location information of objects. Afterward, we fuse these two features and employ ReLU activation, a convolution block, and sigmoid activation to obtain the activation map as follows:

$$F_{e1}^{i-1} = \mathrm{CBR}_1\left(F^{i-1}\right), \quad F_{e2}^{i} = \mathrm{CBR}_2\left(F^{i}\right) \quad (14)$$

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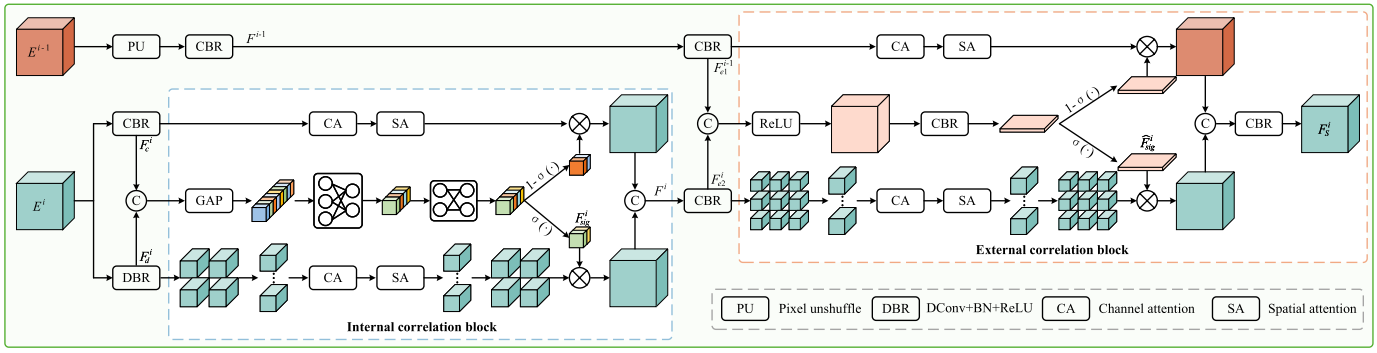


Fig. 4. Illustration of HSRM.

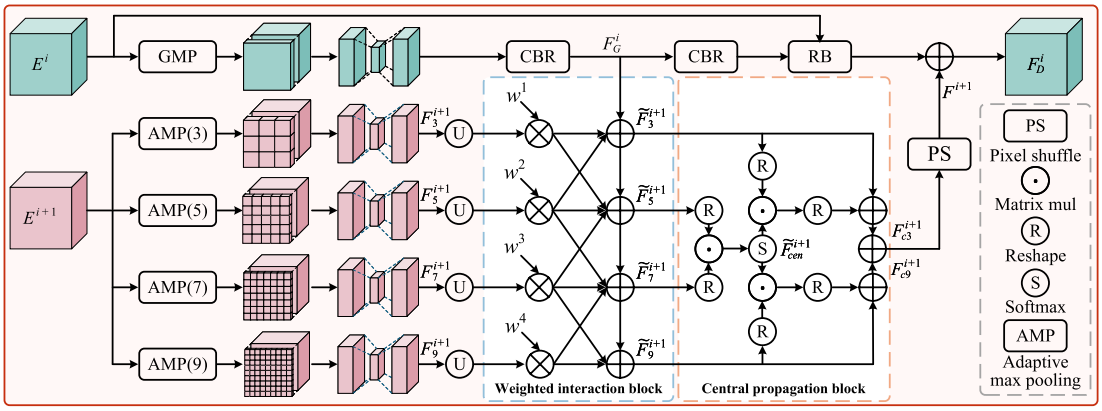


Fig. 5. Illustration of DDIM.

$$\widehat{F}^i_{\text{sig}} = \text{Sigmoid}\left(\text{CBR}\left(\text{ReLU}\left(\text{Cat}\left(F^{i-1}_{e1}, F^i_{e2}\right)\right)\right)\right) \quad (15)$$

where CBR_1 and CBR_2 represent the two convolutional blocks. On this basis, we divide the feature F^i_{e2} into three equal parts along the H- and W-directions to obtain nine smaller feature vectors and perform more elaborate local weighting and recombination in the same way. Similarly, we extract global clues from another high-level feature F^{i-1}_{e1} and correlate and merge these two enhanced high-level features with activation map $\widehat{F}^i_{\text{sig}}$ and its derived map to select representative semantic information between the features. The above procedure can be defined as follows:

$$F^i_S = \text{Cat}\left(\text{CS}\left(F^{i-1}_{e1}\right)\otimes\left(1-\widehat{F}^i_{\text{sig}}\right), \quad \text{RCS}\left(F^i_{e2}\right)\otimes\widehat{F}^i_{\text{sig}}\right) \quad (16)$$

where F^i_S is the output of the entire HSRM.

The proposed HSRM deeply mines representative semantic information within and between features through two progressive feature reconstructions, which significantly enhances the ability of the model to suppress background noise and locate salient regions.

D. Dynamic Detail Interaction Module

The detailed information from low-level encoder features is beneficial for enhancing the discriminability of similar objects and maintaining the integrity of prediction maps. However, these features generally also bring a large amount of noise contamination, which may result in generated images including many blurry regions. To this end, we propose the DDIM to dynamically capture multiscale features and progressively aggregate cross-scale representations.

The structure of DDIM is illustrated in Fig. 5. Different from the direct concatenation of pyramid pooling module (PPM), our DDIM effectively preserves the detailed information of features at different scales through weighted interaction and progressive aggregation. Specifically, given the inputs E^i and E^{i+1} for the ith DDIM, we first use global max pooling on lower level feature E^i to find the region with the greatest visual impact. Then, for higher level feature E^{i+1}, we apply adaptive max pooling at four scales to explore multiscale detailed information. Afterward, we compress and expand the channel information of these five features to reduce computational complexity and enhance nonlinear characteristics, and apply an additional convolutional block on the lower level feature to adjust the channel information, which can be expressed as follows:

$$F^i_G = \text{CBR}\left(\text{CE}\left(\text{GMP}\left(E^i\right)\right)\right) \quad (17)$$

$$F^{i+1}_n = \text{CE}\left(\text{AMP}_n\left(E^{i+1}\right)\right), n = 3, 5, 7, 9 \quad (18)$$

where CE means the compression and expansion operations, and AMP_n represents the adaptive max pooling with the size of $n \times n$. On this basis, to enable the model to autonomously determine the proportion of different scales, we upsample the multiscale feature F^{i+1}_n to the original resolution and input it into the weighted interaction block (WIB). This block equips

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each scale with a learnable parameter. The sum of these four parameters is 1 and initialized to 0.25, respectively. Subsequently, to sufficiently complement information from different scales, we design the fusion mode of adjacent interaction and utilize the global feature F_G^i for progressive guidance to generate a new feature $\widetilde{F}_n^{i+1}$. In addition, different from some existing methods, we propose the central propagation block (CPB) to effectively aggregate information at different scales. Specifically, considering that large and small receptive fields are susceptible to noise interference and background redundancy, we fuse and activate two intermediate scale features to generate the central feature $\widetilde{F}_{\text{cen}}^{i+1}$, which can be expressed as

$$\widetilde{F}_{\text{cen}}^{i+1} = \text{softmax}\left(R\left(\widetilde{F}_5^{i+1}\right) \odot R\left(\widetilde{F}_7^{i+1}\right)\right) \tag{19}$$

where softmax represents the softmax function, R represents the reshape operation, and $\odot$ means the matrix multiplication. It is worth noting that compared with addition or channel concatenation, matrix multiplication and softmax contribute to enhancing the detailed representation of objects without introducing redundant channel information. Afterward, we propagate the central information to surroundings and fuse it with both large- and small-scale feature to obtain the propagation feature F_{cm}^{i+1}. Meanwhile, we further merge feature F_{cm}^{i+1} and apply pixel shuffle [42] to obtain the final higher level feature F^{i+1}. The above process can be described as follows:

$$F_{cm}^{i+1} = R\left(\widetilde{F}_{\text{cen}}^{i+1} \odot R\left(\widetilde{F}_m^{i+1}\right)\right) \oplus \widetilde{F}_m^{i+1} \quad , m = 3, 9 \tag{20}$$

$$F^{i+1} = \text{PS}\left(F_{c3}^{i+1} \oplus F_{c9}^{i+1}\right) \tag{21}$$

where PS is the pixel shuffle operation. Finally, the aggregated multiscale information from the higher layer is used as auxiliary information to supplement the lower layer feature, which can be indicated as

$$F_D^i = \text{RB}\left(\text{CBR}\left(F_G^i\right)\right) \oplus F^{i+1} \tag{22}$$

where F_D^i is the output of the entire DDIM.

The DDIM focuses on the detailed information of the lower layers of encoder. It explores the diverse local texture details of objects through the WIB and enhances the connections between local contexts through the CPB, thereby providing better salient object inference capabilities.

E. Loss Function

During the training process, we use the mixed loss function to supervise the saliency and edge maps output by two decoders, including binary cross-entropy (BCE) loss and intersection over union (IoU) loss. We define the mixed loss function as follows:

$$\mathcal{L}_X = \mathcal{L}_{\text{bce}}(X, G) + \mathcal{L}_{\text{iou}}(X, G) \tag{23}$$

where $\mathcal{L}_{\text{bce}}$ and $\mathcal{L}_{\text{iou}}$ are the BCE loss and IoU loss, respectively, X is the corresponding prediction map, and G is the ground truth. The total loss function $\mathcal{L}_{\text{total}}$ can be formulated as follows:

$$\mathcal{L}_{\text{total}} = \lambda \mathcal{L}_{\text{edg}} + \mu \mathcal{L}_{\text{sal}} \tag{24}$$

where $\mathcal{L}_{\text{edg}}$ is the loss of edge branch, $\mathcal{L}_{\text{sal}}$ is the loss of saliency branch, and λ and μ are set to 0.3 and 1.5 in our experiments to ensure training stability, respectively.

IV. Experiments

A. Experimental Setup

1) *Datasets:* We choose three publicly available datasets, including ORSSD [65], EORSSD [66], and ORSI-4199 [37], to comprehensively evaluate the effectiveness of our model. ORSSD is the earliest public dataset of ORSI-SOD, which contains 600 training images and 200 testing images. The images in this dataset have high resolution and diverse scenes. EORSSD is the extended version of ORSSD, which contains 1400 training images and 600 testing images. Compared to the ORSSD dataset, the images in EORSSD have more scene types and higher complexity. ORSI-4199 is the latest and largest dataset to date, which contains 2000 training images and 2199 testing images. This dataset has higher quality pixel-level annotations and more challenging issues such as complex backgrounds, lighting shadows, and imaging blur.
2) *Implementation Details:* All of our experiments are conducted on a single NVIDIA GeForce RTX 3090 GPU with 24-GB memory. Following most transformer-based methods, we use PVT-v2-b2 [67] as the backbone and crop the input image to 352×352. To avoid overfitting during training, we use flipping and rotation operations to enhance the images in the training set. Our ERNet adopts the Adam optimizer to update model parameters and sets the batch size to 8. The initial learning rate and epoch are set to $1e^{-4}$ and 45. The learning rate decays to one-tenth of its original level after 30 epochs.
3) *Evaluation Metrics:* We employ four widely adopted evaluation metrics to evaluate the performance of our model and other advanced models on three datasets, including S-measure (S_α) [68], F-measure (F_β) [69], E-measure (E_ξ) [70], and mean absolute error ($\mathcal{M}$) [71]. Among them, F-measure and E-measure contain three scores: maximum value, mean value, and adaptive value. In addition to evaluating the saliency map, we also analyze the floating-point operations (FLOPs) and parameter counts of the model to evaluate its deployment effect. Furthermore, we also plot the precision–recall (PR) curves and F-measure curves to more intuitively reflect the performance of models.

B. Comparison With State of the Arts

1) *Comparison Methods:* We conduct a comprehensive comparison between our ERNet and other 30 state-of-the-art methods in the fields of NSI-SOD and ORSI-SOD. The selected competitors comprise four main categories, including CNN-based NSI-SOD methods (i.e., DPORTNet [43] and DNTD [44]), transformer-based NSI-SOD methods (i.e., VST [45] and ICON [46]), CNN-based ORSI-SOD methods (i.e., CorrNet [47], MCCNet [24], ERPNet [10], ACCoNet [40], AESINet [12], SDNet [48], BAFSNet [11], SeaNet [49], SEINet [50], MCPNet [51], LSHNet [52], ADSTNet [53], SFANet [54], MEANet [55], ISAANet [56],

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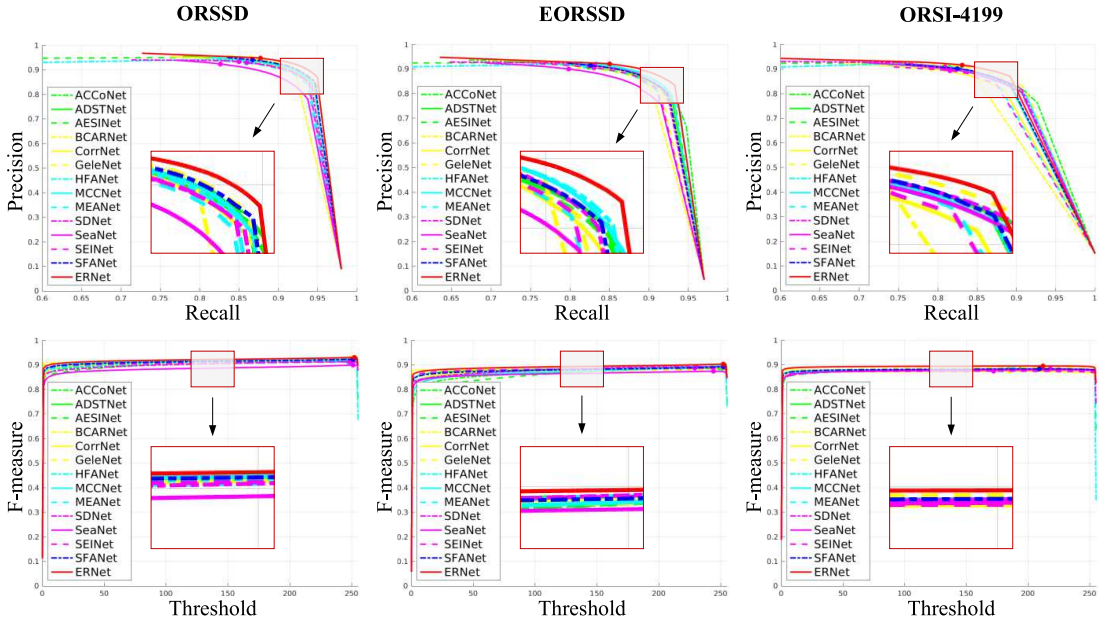


Fig. 6. Comparisons with other methods in terms of PR and F-measure curves on three benchmark datasets.

TABLE I

QUANTITATIVE COMPARISONS BETWEEN THE PROPOSED ERNET AND 30 STATE-OF-THE-ART METHODS ON ORSSD AND EORSSD DATASETS. THESE METHODS CAN BE CLASSIFIED INTO CNN-BASED NSI-SOD (CN), TRANSFORMER-BASED NSI-SOD (TN), ORDINARY OR LIGHTWEIGHT CNN-BASED ORSI-SOD (OCO/LCO), AND TRANSFORMER-BASED ORSI-SOD (TO). ↑ AND ↓ MEAN THAT THE HIGHER AND LOWER IS BETTER. THE TOP TWO RESULTS ARE MARKED IN RED AND BLUE, RESPECTIVELY

Methods	Type	Backbone	FLOPs $(G)\downarrow$	Params $(M)\downarrow$	ORSSD								EORSSD							
					$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{adp}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{adp}\uparrow$	$\mathcal{M}\downarrow$
DPORTNet$_{22}$ [43]	CN	ResNet	61	19	.8827	.8309	.8184	.7970	.9214	.9139	.9083	.0220	.8960	.8363	.7937	.7545	.9423	.9116	.9150	.0150
DNTD$_{22}$ [44]	CN	ResNet	10.9	28.8	.8698	.8231	.8020	.7645	.9286	.9086	.9081	.0217	.8957	.8189	.7962	.9288	.9378	.9225	.9047	.0113
VST$_{21}$ [45]	TN	T2T-ViT	23.2	44	.9365	.9095	.8817	.8262	.9810	.9621	.9466	.0094	.9208	.8716	.8263	.7089	.9743	.9442	.8941	.0067
ICON$_{23}$ [46]	TN	PVT	21	33.1	.9256	.8939	.8671	.8444	.9704	.9637	.9554	.0116	.9185	.8622	.8371	.8065	.9687	.9619	.9497	.0073
CorrNet$_{22}$ [47]	LCO	VGG	21.1	4.1	.9380	.9129	.9002	.8875	.9790	.9746	.9721	.0098	.9289	.8778	.8620	.8311	.9696	.9646	.9593	.0083
MCCNet$_{22}$ [24]	OCO	VGG	117	67.7	.9437	.9155	.9054	.8957	.9800	.9758	.9735	.0087	.9327	.8904	.8604	.8137	.9755	.9685	.9538	.0066
ERPNet$_{23}$ [10]	OCO	ResNet	87.2	56.5	.9254	.8974	.8745	.8356	.9710	.9566	.9520	.0135	.9210	.8632	.8304	.7554	.9603	.9401	.9228	.0089
ACCoNet$_{23}$ [40]	OCO	VGG	368.6	102.6	.9437	.9149	.8971	.8806	.9796	.9754	.9721	.0088	.9290	.8837	.8552	.7969	.9727	.9653	.9450	.0074
AESINet$_{23}$ [12]	OCO	ResNet	23.4	51.6	.9455	.9160	.8962	.8640	.9814	.9741	.9689	.0085	.9347	.8792	.8496	.7909	.9757	.9647	.9480	.0064
SDNet$_{23}$ [48]	OCO	ResNet	-	-	.9340	.9074	.8932	.8752	.9768	.9717	.9677	.0101	.9359	.8853	.8680	.8370	.8704	.8644	.9607	.0065
BAFSNet$_{23}$ [11]	OCO	ResNet	16	31	.9385	.9136	.9017	.8982	.9819	.9785	.9779	.0079	.9286	.8708	.8583	.8477	.9772	.9735	.9729	.0060
SeaNet$_{23}$ [49]	LCO	MobileNet	1.7	2.76	.9260	.8942	.8772	.8625	.9767	.9722	.9670	.0105	.9208	.8649	.8519	.8304	.9710	.9651	.9602	.0073
SEINet$_{23}$ [50]	OCO	VGG	50.1	20.2	.9382	.9090	.8996	.8946	.9784	.9746	.9763	.0097	.9283	.8788	.8708	.8587	.8723	.9672	.9657	.0076
MCPNet$_{24}$ [51]	OCO	VGG	86.6	25.1	.9433	.9135	.8979	.8726	.9807	.9736	.9724	.0090	.9373	.8868	.8620	.8077	.9765	.9657	.9515	.0070
LSHNet$_{24}$ [52]	OCO	ResNet	-	-	.9491	.9200	.9054	.8848	.9824	.9764	.9758	.0075	.9370	.8844	.8643	.8261	.9761	.9692	.9619	.0064
ADSTNet$_{24}$ [53]	OCO	ResNet	27.7	62.1	.9379	.9124	.9042	.8979	.9807	.9740	.9785	.0086	.9311	.8804	.8716	.8532	.9769	.9709	.9681	.0065
SFANet$_{24}$ [54]	OCO	ResNet	7.7	25.1	.9453	.9192	.9063	.8984	.9830	.9789	.9800	.0077	.9349	.8833	.8680	.8492	.9769	.9726	.9681	.0058
MEANet$_{24}$ [55]	LCO	MobileNet	9.6	3.3	.9340	.9033	.8934	.8860	.9768	.9730	.9715	.0098	.9282	.8766	.8678	.8557	.9708	.9658	.9663	.0070
ISAANet$_{24}$ [56]	LCO	MobileNet	4.3	3.6	.9429	.9166	.9034	.8982	.9836	.9787	.9777	.0077	.9351	.8869	.8748	.8643	.9761	.9717	.9734	.0063
SOLNet$_{25}$ [57]	LCO	RepVGG	8.1	6.5	.9284	.9012	.8946	.8925	.9734	.9689	.9697	.0111	.9171	.8609	.8513	.8392	.9623	.9577	.9596	.0078
BCARNet$_{25}$ [58]	OCO	ResNet	7	24	.9466	.9196	.9070	.9073	.9833	.9799	.9796	.0071	.9360	.8871	.8736	.8740	.9761	.9728	.9734	.0054
EANet$_{25}$ [59]	OCO	ResNet	30.1	25.2	.9310	.9180	-	-	.9759	-	-	.0113	.9286	.8974	-	-	.9740	-	-	.0073
MRBINet$_{25}$ [60]	OCO	ResNet	42.8	32.4	.9474	.9199	.9110	.9058	.9851	.9812	.9793	.0069	.9351	.8852	.8768	.8626	.9766	.9721	.9729	.0056
HFANet$_{22}$ [36]	TO	PVT	68.3	60.5	.9399	.9112	.8981	.8819	.9770	.9712	.9722	.0092	.9380	.8876	.8681	.8365	.9740	.9679	.9644	.0070
ASTTNet$_{23}$ [28]	TO	T2T-ViT	44.6	13	.9348	.9061	.8811	.8456	.9795	.9699	.9587	.0094	.9253	.8741	.8297	.7534	.9757	.9584	.9247	.0059
GeleNet$_{23}$ [27]	TO	SwinT	11.7	25.5	.9410	.9203	.9093	.9038	.9829	.9779	.9805	.0080	.9259	.8774	.8649	.8528	.9794	.9752	.9713	.0055
GLGCNet$_{23}$ [61]	TO	PVT	9.8	25.2	.9488	.9236	.9054	.8931	.9864	.9820	.9800	.0071	.9375	.8924	.8714	.8499	.9803	.9757	.9701	.0055
WeightNet$_{24}$ [62]	TO	PVT	21.2	66.3	.9512	.9278	.9125	.8979	.9864	.9823	.9776	.0075	.9417	.8908	.8779	.8521	.9809	.9749	.9670	.0060
ASNet$_{24}$ [63]	TO	PVT	20.1	65.4	.9441	.9172	.9072	.8986	.9803	.9764	.9795	.0081	.9345	.8959	.8770	.8672	.9783	.9745	.9748	.0055
LGIPNet$_{25}$ [64]	TO	PVT	47	52.1	.9507	.9266	.9129	.9027	.9876	.9829	.9809	.0066	.9392	.8924	.8770	.8623	.9806	.9753	.9719	.0053
ERNet (ours)	TO	PVT	11.9	67.1	.9525	.9287	.9152	.9080	.9872	.9832	.9814	.0069	.9430	.8975	.8839	.8697	.9842	.9791	.9767	.0049

SOLNet [57], BCARNet [58], EANet [59], and MRBINet [60]), and transformer-based ORSI-SOD methods (i.e., HFANet [36], ASTTNet [28], GeleNet [27], GLGCNet [61], WeightNet [62], ASNet [63], and LGIPNet [64]). Notably, since ORSI-4199 is the latest released dataset, some models do not provide results for

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TABLE II

COMPARISON WITH OTHER TRANSFORMER-BASED METHODS ON INFERENCE SPEED

Model	$HFANet_{22}$ [36]	$ASTTNet_{23}$ [28]	$GeleNet_{23}$ [27]	$GLGCNet_{23}$ [61]
FPS	26	13	29	21
Model	$WeightNet_{24}$ [62]	$ASNet_{24}$ [63]	$LGIPNet_{25}$ [64]	ERNet (ours)
FPS	40	61	36	31

this dataset. Therefore, we select some representative methods for comparison.

2) *Quantitative Comparison:* As shown in Table I, we present a quantitative comparison with the advanced model using eight evaluation metrics on the ORSSD and EORSSD datasets. The data show that transformer-based methods outperform CNN-based methods in most cases in terms of NSI-SOD and ORSI-SOD. As the transformer-based method, our ERNet is leading in the vast majority of metrics. Specifically, in the ORSSD dataset, ERNet outperforms other competing methods in six metrics. For example, compared to the second-place LGIPNet, our method achieves 0.23% and 0.53% improvements in the F_β^{mean} and F_β^{adp} metrics. In the EORSSD dataset, ERNet ranks first among seven metrics. It is worth noting that the metric F_β^{adp} of ERNet falls behind that of BCARNet on the EORSSD dataset may be because BCARNet designs the attention restoration mechanism to compensate for the lack of local attention and suppress the interference of background noise. Notably, among all the evaluated models, ERNet is the only one with the $\mathcal{M}$ metric lower than 0.007 and 0.005 for the ORSSD and EORSSD datasets, respectively. The left two columns of Fig. 6 further display the PR and F-measure curves of all compared models on the ORSSD and EORSSD datasets. It can be seen that the PR curves of our method on the two datasets are significantly closer to the top-right corner, and the F-measure curves are remarkably higher than the other methods. Moreover, in mainstream models, our ERNet achieves an excellent balance between model performance and computational efficiency through lower FLOPs and parameter counts. More importantly, Table II indicates that the inference speed of the proposed ERNet also has some advantages over the compared transformer-based methods.

On the most challenging ORSI-4199 dataset, our method also obtains satisfactory results. As shown in Table III, it can be seen that our ERNet achieves the best performance in six metrics and the second-best performance in two metrics. When compared to GeleNet, ERNet significantly improves 0.36% and 0.8% in key metrics S_α and F_β^{mean}. Similarly, in comparison with GLGCNet, our ERNet outperforms 0.53%, 1.02%, and 0.37% in terms of $F_\beta^{\max}$, F_β^{mean}, and F_β^{adp}, respectively. The PR and F-measure curves shown in the rightmost column of Fig. 6 provide additional verification of our superiority over other state-of-the-art models.

TABLE III

FURTHER QUANTITATIVE COMPARISONS BETWEEN THE PROPOSED ERNET AND 24 REPRESENTATIVE METHODS ON THE ORSI-4199 DATASET. ↑ AND ↓ MEAN THAT THE HIGHER AND LOWER IS BETTER. THE TOP TWO RESULTS ARE MARKED IN RED AND BLUE, RESPECTIVELY

Methods	Type	ORSI-4199							
		$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$F_\beta^{mean} \uparrow$	$F_\beta^{adp} \uparrow$	$E_\xi^{max} \uparrow$	$E_\xi^{mean} \uparrow$	$E_\xi^{adp} \uparrow$	$\mathcal{M} \downarrow$
$DPORTNet_{22}$ [43]	CN	.8094	.7789	.7701	.7554	.8759	.8687	.8628	.0569
$DNTD_{22}$ [44]	CN	.8444	.8310	.8208	.8065	.9158	.9050	.8963	.0425
VST_{21} [45]	TN	.8790	.8717	.8524	.7947	.9481	.9348	.8997	.0281
$ICON_{23}$ [46]	TN	.8752	.8763	.8664	.8531	.9521	.9438	.9239	.0282
$CorrNet_{22}$ [47]	LCO	.8623	.8560	.8513	.8534	.9330	.9206	.9142	.0366
$MCCNet_{22}$ [24]	OCO	.8746	.8690	.8630	.8592	.9413	.9348	.9182	.0316
$ERPNet_{23}$ [10]	OCO	.8670	.8553	.8374	.8024	.9290	.9149	.9024	.0357
$ACCoNet_{23}$ [40]	OCO	.8761	.8657	.8591	.8562	.9398	.9297	.9163	.0314
$AESINet_{23}$ [12]	OCO	.8657	.8467	.8372	.8072	.9399	.9311	.9200	.0333
$SDNet_{23}$ [48]	OCO	.8778	.8625	.8534	.8295	.9445	.9361	.9157	.0316
$SeaNet_{23}$ [49]	LCO	.8722	.8653	.8591	.8556	.9426	.9363	.9197	.0308
$SEINet_{23}$ [50]	OCO	.8686	.8616	.8572	.8571	.9369	.9305	.9138	.0330
$ADSTNet_{24}$ [53]	OCO	.8710	.8698	.8653	.8655	.9433	.9356	.9212	.0318
$SFANet_{24}$ [54]	OCO	.8761	.8710	.8659	.8647	.9447	.9385	.9202	.0292
$MEANet_{24}$ [55]	LCO	.8708	.8662	.8624	.8638	.9417	.9338	.9207	.0312
$ISAANet_{24}$ [56]	LCO	.8732	.8760	.8696	.8694	.9497	.9431	.9239	.0281
$BCARNet_{25}$ [58]	OCO	.8757	.8689	.8651	.8666	.9407	.9365	.9190	.0306
$EANet_{25}$ [59]	OCO	.8683	.8563	-	-	.9287	-	-	.0361
$MRBINet_{25}$ [60]	OCO	.8824	.8800	.8755	.8755	.9489	.9425	.9220	.0268
$HFANet_{22}$ [36]	TO	.8767	.8700	.8624	.8323	.9431	.9336	.9191	.0314
$ASTT_{23}$ [28]	TO	.8828	.8788	.8673	.8527	.9512	.9435	.9216	.0273
$GeleNet_{23}$ [27]	TO	.8828	.8806	.8734	.8681	.9537	.9482	.9261	.0264
$GLGCNet_{23}$ [61]	TO	.8839	.8808	.8712	.8672	.9508	.9469	.9245	.0274
$LGIPNet_{25}$ [64]	TO	.8854	.8791	.8668	.8268	.9499	.9432	.9213	.0276
ERNet (ours)	TO	.8864	.8861	.8814	.8709	.9542	.9471	.9295	.0271

3) *Qualitative Comparison:* In Fig. 7, we exhibit five representative and challenging ORSI scenarios for qualitative comparison between ERNet and competing methods, including bridge, car, building, airplane, and ship. Specifically, in terms of bridge detection, the competing methods struggle to completely segment the bridge when they are disturbed by rivers and coasts. Compared to competitive methods, our approach can generate object boundaries with high accuracy and suppress interference. In the detection of cars, ERNet can correctly detect multiple small cars and capture complex details. In contrast, competitive methods such as BCARNet and SDNet perform poorly in object localization, which leads to serious missed detections. In the detection of buildings and airplanes, many competing methods have obvious misdetection issues, such as the roads between buildings and the houses next to airplanes. Fortunately, our ERNet can effectively suppress confounding interference from similar or adjacent objects, which contributes to distinguishing the boundaries between salient objects and interfering backgrounds. In the detection task of ships, detecting ships in low-light environments is a challenging problem for existing methods. As shown in the 14th row of Fig. 7, our ERNet overcomes the difficulty caused by low contrast between foreground and background and precisely detects salient objects from the background.

4) *Attribute-Based Analysis on ORSI-4199:* In the ORSI-4199 dataset, images can be further divided into nine

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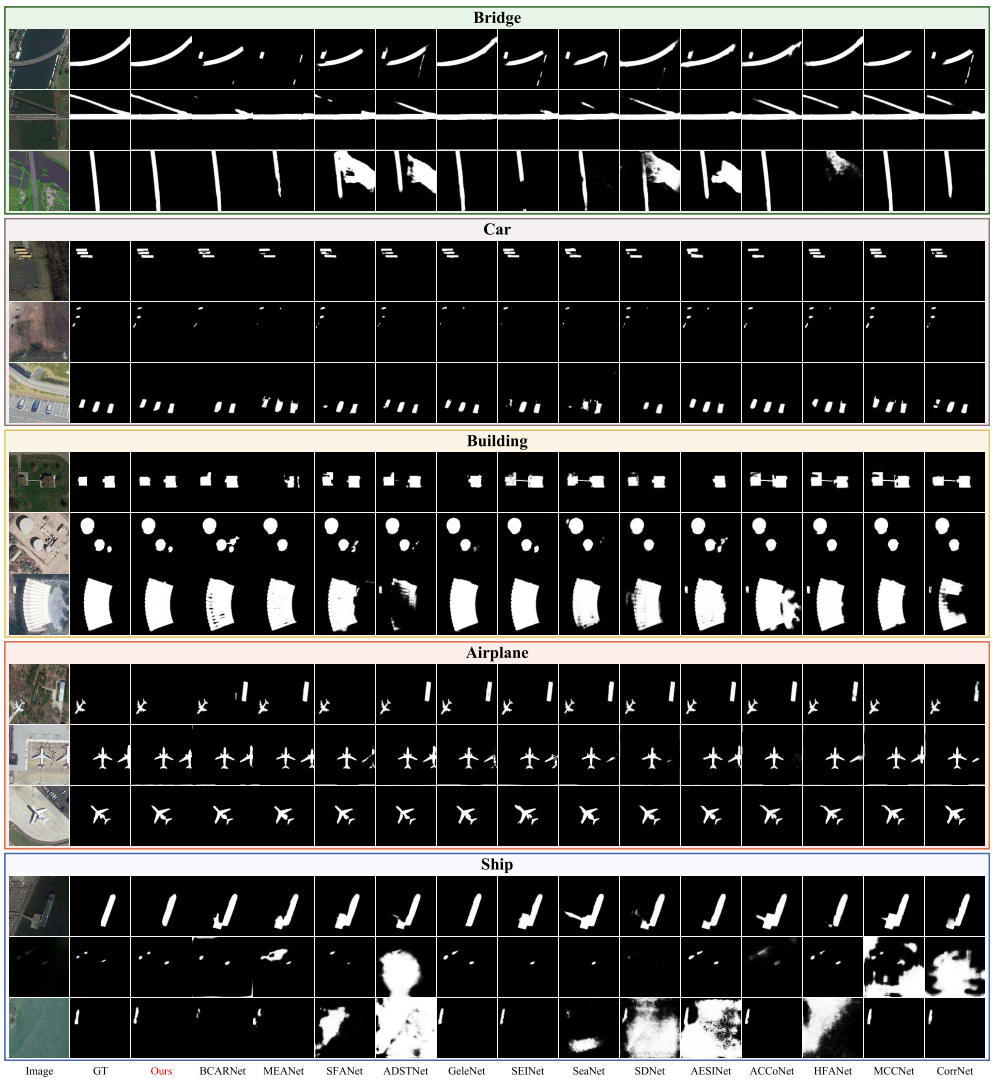


Fig. 7. Qualitative comparisons of our ERNet and other methods.

small datasets based on nine challenging attribute scenes. Table IV shows the further performance evaluation of the proposed model and several state-of-the-art models on nine attributes. From the table, we can see that our model ranks first on seven attributes and second on one attribute. Furthermore, our model also achieves the best result in terms of average performance on nine attributes. Particularly, our model significantly outperforms other models in handling multiple salient objects (MSOs) and small salient object (SSO), demonstrating the strong ability in locating salient objects.

TABLE IV

ATTRIBUTE-BASED PERFORMANCE ANALYSIS ON THE ORSI-4199 DATASET. THE LAST ROW PRESENTS THE AVERAGE RESULTS FOR THE NINE ATTRIBUTES. THE TOP TWO MAX F_β ARE MARKED IN RED AND BLUE, RESPECTIVELY. FOR THE COMPLETE EXPLANATION OF EACH ATTRIBUTE, REFER TO [37]

Attr	HFANet [36]	CorrNet [47]	ACCoNet [40]	AESINet [12]	ASTT [28]	ADSTNet [53]	Ours
BSO	.9103	.8891	.9091	.9109	.9180	.9180	.9274
CS	.8964	.8835	.8974	.9068	.9090	.9012	.9151
CSO	.8948	.8728	.8931	.9001	.8900	.8940	.8936
ISO	.9046	.8742	.8997	.9093	.9230	.8995	.9168
LCO	.8030	.7852	.8048	.8136	.7850	.8092	.8207
MSO	.8566	.8528	.8611	.8579	.8570	.8567	.8765
NSO	.8858	.8724	.8814	.8922	.8980	.8895	.9059
OC	.8506	.8409	.8337	.8349	.8370	.8503	.8643
SSO	.8128	.8051	.8065	.8079	.8000	.8056	.8332
Avg	.8682	.8529	.8652	.8704	.8690	.8693	.8837

C. Ablation Studies

In this section, we conduct a series of ablation experiments to validate the effectiveness of the three modules proposed in ERNet, including ablation experiments with different combinations between modules and ablation experiments within each module.

1) *Ablation Studies Between Different Modules:* To explore the individual and combined contributions of the three proposed modules to ERNet, we design eight variant models, and the results are summarized in Table V. From the table, we can see that the model performance is significantly higher than the baseline model after sequentially integrating AUCM, HSRM, and DDIM, which directly proves the effectiveness of the three

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TABLE V

ABLATION STUDY OF OUR PROPOSED MODULES. BASE. MEANS BASELINE. THE BEST RESULTS ARE IN BOLD

No.	Base.	Modules			ORSSD								EORSSD								Complexity	
		AUCM	HSRM	DDIM	$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{adp}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{adp}\uparrow$	$\mathcal{M}\downarrow$	FLOPs(G)	Params(M)
1	✓				.9332	.9017	.8871	.8767	.9835	.9774	.9780	.0106	.9201	.8584	.8408	.8185	.9773	.9707	.9616	.0082	4.3	26.3
2	✓	✓			.9450	.9204	.9072	.8976	.9837	.9790	.9795	.0085	.9380	.8908	.8708	.8505	.9802	.9724	.9682	.0063	8.1	29.8
3	✓		✓		.9372	.9082	.8942	.8860	.9833	.9787	.9794	.0088	.9305	.8787	.8640	.8476	.9792	.9739	.9701	.0067	7.3	60.4
4	✓			✓	.9403	.9142	.9009	.8922	.9807	.9755	.9765	.0095	.9330	.8830	.8666	.8441	.9760	.9700	.9662	.0068	5.1	29.4
5	✓	✓	✓		.9466	.9195	.9047	.8910	.9852	.9805	.9787	.0075	.9383	.8882	.8753	.8617	.9789	.9739	.9705	.0058	11.1	64.0
6	✓	✓		✓	.9482	.9214	.9103	.9027	.9854	.9811	.9819	.0079	.9405	.8938	.8804	.8664	.9811	.9762	.9738	.0055	8.9	32.9
7	✓		✓	✓	.9442	.9214	.9075	.8989	.9826	.9781	.9792	.0079	.9371	.8889	.8732	.8570	.9803	.9753	.9736	.0056	8.1	63.6
8	✓	✓	✓	✓	**.9525**	**.9287**	**.9152**	**.9080**	**.9872**	**.9832**	**.9814**	**.0069**	**.9430**	**.8975**	**.8839**	**.8697**	**.9842**	**.9791**	**.9767**	**.0049**	11.9	67.1

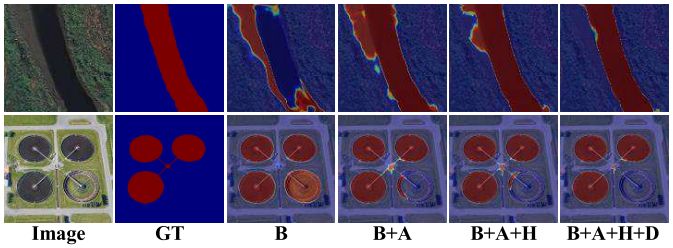


Fig. 8. Visual comparison of different proposed modules in ablation experiments. B, B + A, B + A + H, and B + A + H + D denote baseline, baseline + AUCM, baseline + AUCM + HSRM, and baseline + AUCM + HSRM + DDIM, respectively.

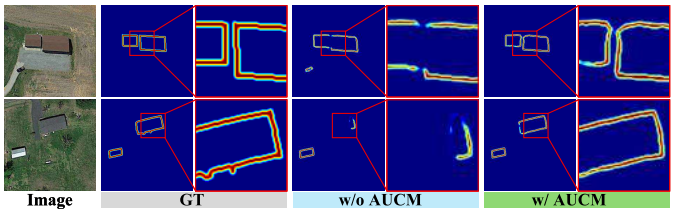


Fig. 9. Visualization results of AUCM in the edge region.

TABLE VI

ABLATION STUDY OF DIFFERENT COMPONENTS IN AUCM. THE BEST RESULTS ARE IN BOLD

Schemes	EORSSD							
	$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{adp}\uparrow$	$\mathcal{M}\downarrow$
w/o DBG	.9401	.8921	.8766	.8608	.9792	.9741	.9712	.0057
w/o SEPM	.9375	.8917	.8766	.8575	.9794	.9744	.9709	.0059
r/ DP	.9361	.8869	.8713	.8561	.9781	.9737	.9701	.0058
w/ DB and SEPM	**.9430**	**.8975**	**.8839**	**.8697**	**.9842**	**.9791**	**.9767**	**.0049**

TABLE VII

ABLATION STUDY OF DIFFERENT SIMILARITY CALCULATION SCALES IN SEPM. THE BEST RESULTS ARE IN BOLD

Scale	EORSSD							
	$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{adp}\uparrow$	$\mathcal{M}\downarrow$
$(5\times5, 3\times3)$	.9413	.8965	.8827	.8673	.9838	.9784	.9731	.0052
$(7\times7, 3\times3)$	.9412	.8942	.8789	.8644	.9803	.9752	.9711	.0057
$(9\times9, 3\times3)$	.9403	.8921	.8797	.8669	.9805	.9758	.9740	.0059
$(9\times9, 5\times5)$	**.9430**	**.8975**	**.8839**	**.8697**	**.9842**	**.9791**	**.9767**	**.0049**
$(9\times9, 7\times7)$	.9400	.8926	.8802	.8659	.9817	.9768	.9736	.0062

proposed modules. Specifically, compared to the first four rows of the table, the introduction of these modules results in a 1.79%, 1.04%, and 1.29% increase in the S_α metric and the 0.19%, 0.15%, and 0.14% decrease in the $\mathcal{M}$ metric for the EORSSD dataset, respectively. Meanwhile, the combination of these three modules also brings considerable performance improvements, which can be seen in the fifth–seventh rows of the table. Compared to the baseline model, ERNet with all the modules brings the substantial improvement of 2.29%, 3.91%, 4.31%, and 5.12% on the four metrics S_α, $F_\beta^{\max}$, F_β^{mean}, and F_β^{adp} for the EORSSD dataset. In addition, Fig. 8 provides the visual examples of proposed modules. It can be seen that adding AUCM to the baseline can significantly improve the prediction results, and the model detection performance is optimal when all modules are employed simultaneously.

2) *Ablation Studies Inside the AUCM Module:* To investigate the effectiveness of crucial components in AUCM, we provide three additional variant modules of AUCM: 1) removing the guidance of the global average pooling and global max pooling branches (i.e., w/o DBG); 2) removing SEPM (i.e., w/o SEPM); and 3) replacing SEPM with difference perception (DP) in AESINet [12] (i.e., r/ DP). The experimental results are shown in Table VI. It can be found that removing either the DBG or SEPM has a negative impact on performance, which indicates that the fusion of edge and global information contributes to the improvement of model performance. Meanwhile, the model performance shows a significant decrease when the simple DP module is used instead of the SEPM module. We can conclude that our proposed SEPM can more effectively perceive edge information between foreground and background. Furthermore, we also investigate the influence of similarity computation at different scales within SEPM on model performance. As shown in Table VII, we set the first and second similarity calculation sizes to the minimum of 5×5 and 3×3, respectively. By gradually increasing these two sizes, we can see that the model achieves the best performance when these two sizes are set to 9×9 and 5×5. In addition, Fig. 9 provides the visual comparison of edge region before and after the introduction of AUCM. It can be seen that salient objects contain more complete edges after the edge guidance of AUCM.

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TABLE VIII

ABLATION STUDY OF DIFFERENT PARTS IN HSRM. THE BEST RESULTS ARE IN BOLD

Schemes	EORSSD							
	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$F_\beta^{mean} \uparrow$	$F_\beta^{adp} \uparrow$	$E_\xi^{max} \uparrow$	$E_\xi^{mean} \uparrow$	$E_\xi^{adp} \uparrow$	$\mathcal{M} \downarrow$
w/o ICB	.9414	.8949	.8791	.8608	.9815	.9761	.9724	.0054
w/o ECB	.9391	.8931	.8748	.8581	.9810	.9763	.9744	.0055
w/ ICB and ECB	**.9430**	**.8975**	**.8839**	**.8697**	**.9842**	**.9791**	**.9767**	**.0049**

TABLE IX

ABLATION STUDY OF DIFFERENT PARTS IN DDIM. THE BEST RESULTS ARE IN BOLD

Schemes	EORSSD							
	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$F_\beta^{mean} \uparrow$	$F_\beta^{adp} \uparrow$	$E_\xi^{max} \uparrow$	$E_\xi^{mean} \uparrow$	$E_\xi^{adp} \uparrow$	$\mathcal{M} \downarrow$
w/o WIB	.9412	.8949	.8788	.8611	.9817	.9761	.9723	.0055
w/o CPB	.9400	.8913	.8762	.8615	.9797	.9745	.9710	.0057
r/ PPM	.9407	.8910	.8790	.8675	.9808	.9760	.9746	.0057
r/ ASPP	.9390	.8910	.8767	.8632	.9806	.9761	.9745	.0055
w/ WIB and CPB	**.9430**	**.8975**	**.8839**	**.8697**	**.9842**	**.9791**	**.9767**	**.0049**

3) *Ablation Studies Inside the HSRM Module:* To verify the effectiveness of ICB and ECB, we train our net work by sequentially removing these two blocks from HSRM. The results are presented in Table VIII. We observe that the lack of ICB or ECB in HSRM leads to significant performance degradation on all metrics, which demonstrates that these two blocks complement each other by enhancing the internal semantic consistency of features and modeling the external discriminative differences between features. It is worth noting that removing the ECB leads to more performance degradation than removing the ICB, which indicates that interfeature interaction is more crucial for HSRM.
4) *Ablation Studies Inside the DDIM Module:* The DDIM is crucial for supplementing detailed information in the model. To verify the effectiveness of essential components in DDIM, we provide two additional variants: removing the WIB (i.e., w/o WIB) and the CPB (i.e., w/o CPB). The experimental results are reported in Table IX. It can be seen that DDIM without these two components shows significant performance degradation in all metrics, which illustrates the importance of weighted interaction and effective aggregation among branches. The weighted interaction and effective aggregation of spatial information at different scales are beneficial for eliminating mutual interference among different elements, realizing effective integration of information at different scales. In addition, the performance of model significantly decreases when we replace DDIM with two other multiscale modules PPM and atrous spatial pyramid pooling (ASPP), proving that the proposed DDIM can more effectively perceive multiscale detailed information.
5) *Ablation Studies on Different Backbones:* We employ three mainstream backbones, including VGG, ResNet, and SwinT, to evaluate the sensitivity of ERNet to different backbones. As shown in Table X, our ERNet exhibits significant stability on all backbones, demonstrating its excellent flexibility and versatility in different configurations. Since ERNet achieves the best overall performance with lower parameter and computational complexity on PVT, we choose PVT as our default backbone.

TABLE X

ABLATION STUDY OF OUR ERNET ON DIFFERENT BACKBONES. THE BEST RESULTS ARE IN BOLD

Models	EORSSD							
	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$F_\beta^{mean} \uparrow$	$F_\beta^{adp} \uparrow$	$E_\xi^{max} \uparrow$	$E_\xi^{mean} \uparrow$	$E_\xi^{adp} \uparrow$	$\mathcal{M} \downarrow$
ERNet-VGG	.9368	.8898	.8759	.8548	.9780	.9721	.9714	.0067
ERNet-ResNet	.9370	.8872	.8723	.8548	.9770	.9720	.9688	.0059
ERNet-SwinT	.9420	.8956	.8826	.8695	.9833	**.9794**	.9759	**.0048**
ERNet-PVT	**.9430**	**.8975**	**.8839**	**.8697**	**.9842**	.9791	**.9767**	.0049

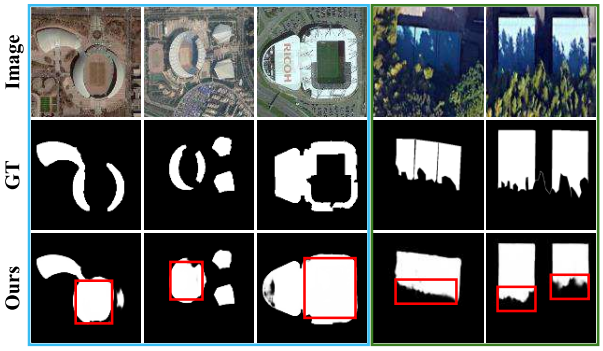


Fig. 10. Failure cases of the proposed ERNet on the challenging scenes.

D. Failure Cases

We provide some failure cases of the proposed ERNet in Fig. 10. Our model has room for further improvement in two types of challenging scenes.

As shown in the first three columns of the figure, it is challenging to completely distinguish salient objects containing nonsalient areas, such as stadiums with internal venues. The main reason is that there are few samples of hollow buildings in the training dataset, which easily leads to deviations in the detection process of model. Moreover, as shown in the last two columns of the figure, accurately detecting salient objects with large areas of shadows is difficult. This is because the salient regions under shadow usually exhibit weak semantic correlation, which is hard for model to capture. Importantly, the extremely challenging scenes mentioned above are also common bottlenecks for most existing methods.

V. CONCLUSION

In this article, we propose a novel ERNet for SOD in ORSIs. Considering the uncertainty of edge regions, we propose the AUCM, which adaptively distinguishes the boundaries of salient objects from coarse to fine through two-stage pixel similarity calculation. For the semantic and detailed information in the encoder, we propose the HSRM and DDIM. The former explores the consistent semantic information of high-level encoder features from both internal and external perspectives, while the latter eliminates the imbalance of

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multiscale detailed information via a WIB and CPB. Extensive comparisons and ablation studies strongly demonstrate that our ERNet outperforms 30 existing state-of-the-art methods on three ORSI-SOD datasets. In the future, our new research focuses on exploring more lightweight and real-time solutions with satisfactory performance. Moreover, borrowing the graph convolution technique to model semantic long-range dependencies also deserves in-depth investigation.

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